

TECHNICAL SPECIFICATIONS FOR X-LINKED POLYETHYLENE INSULATED LT. (1.1KV) AERIAL BUNCHED CABLES

1.0 SCOPE:

This specification covers design, manufacture, testing at manufacturer's works before despatch, transportation, supply and delivery of ISI marked 1100 V grade XLPE insulated, LT Aerial Bunched Cables for overhead lines.

2.0 LOCATION

Aerial bunched Cables for use in L.T. overhead lines will be used in the State of West Bengal under WBSEDCL area.

3.0 CLIMATIC AND ISOCERAUNIC CONDITIONS:

Maximum ambient temperature - 50°C

Minimum ambient temperature - 4°C

Thermal resistance of soil - 1500C –cm/watt

Maximum daily average ambient temperature: 45°C

Maximum relative humidity- 100%

Average rainfall per annum-200 cm

Maximum height above the sea level -1000 meter

Maximum wind pressure/wind speed: 150 kg/sq. m/50m/sec

Average no. of thunderstorm days per annum: 200

4.0 COMPOSITION OF THE CABLE:

The composite cable shall comprise three single-core Power cables with/without one street lighting core twisted over an insulated messenger wire for use as overhead distribution feeders, suitable for use on 3-Phase AC System conforming to IS-14255/1995 with up to date amendments.

Each single-core Power cables should be XLPE insulated stranded compacted circular aluminium conductors.

Street lighting core (if any) should be XLPE insulated stranded compacted circular aluminium conductor.

Messenger Conductor should be stranded compacted circular aluminium alloy conductor & will carry the weight of the cable.

5.0 APPLICABLE STANDARDS:

Unless otherwise stipulated in this specification, the following standard with latest amendment shall be applicable:

IS 14255-1995 and its amendments	Specification for Aerial Bunched Cables for working Voltage up to & including 1100 Volts.
IS : 8130-1984 and its amendments.	Specification for Conductors for Insulated Electric Cables and flexible Cords.
IS : 398 (Part-IV)-1994 and its amendments.	Specification for Aluminium Conductors for overhead transmission purposes: Aluminium Alloy Stranded Conductors (Aluminium-Magnesium –Silicon type).
IS:10418:1982 and its amendments	Specification for Drums for Electric Cables

6.0 RATED VOLTAGE:

The rated voltage of the cables shall be 1100 V.

7.0 DESCRIPTION OF THE CABLE

Description		
1100V Grade L.T. Aerial Bunched Cable with 3(three) Power Cores (Black coloured XLPE insulated Stranded, compacted Circular Aluminium Conductor, Core identification by one, two and three ridges) and one Neutral-cum-Messenger Core (Black coloured XLPE insulated stranded compacted circular Aluminium Alloy wire, Core identification by four ridges), with/without one street lighting core (XLPE insulated stranded compacted circular aluminium conductor without any identification mark) conforming to IS: 8130-1984, IS: 398(Part-IV)-1994 and IS:14255-1995 with latest amendments, if any. (Provision for street light conductor is optional and as per requirement)	Sl. No.	Designation
	1.	LT AB Cable of size 3x16+1x25 sq. mm
	2.	LT AB Cable of size 3x25+1x25 sq. mm.
	3.	LT AB Cable of size 3x35+1x25 sq. mm
	4.	LT AB Cable of size 3X50+1X35 sq. mm.
	5.	LT AB Cable of size 3X50+1X16+1X35 sq. mm.
	6.	LT AB Cable of size 3X70+1X16+1X50 sq. mm
	7.	LT AB Cable of size 3x95+1x16+1x70 sq. mm

8.0 GENERAL TECHNICAL REQUIREMENTS:

8.1 DETAILS OF PHASE/STREET LIGHT CONDUCTORS

The Aluminium Conductors shall be H4 grade class 2 complying with the requirements as specified in IS-8130-1984 **with latest amendments**. The phase and street light conductors shall be of multi stranded aluminium of compacted circular cross section. The aluminium shall comply with IS: 8130 with its amendments. The conductor shall be clean & reasonably uniform in size and shape and its surface shall be free from sharp edges.

8.1.1 INSULATION OF XLPE

The Conductor (with protective barrier, wherever applied) shall be provided with **cross linked polyethylene (XLPE) curing with Sioplus technology** conforming to the standard Specification as per IS: 14255/95 **with latest amendment**, when the cables are suitable for use where the combination of ambient temperature and temperature rise due to load, including temperature on exposure to direct sunlight results in conductor temperature not exceeding the following:

Normal continuous Operation	Short circuit Operation
90°C	250°C

The insulation shall be so applied that it fits closely on the conductor (or barrier, if any) and it shall be possible to remove it without damaging the conductor.

Average thickness of insulation shall not be less than the nominal value shown below in tabular form:

Nominal are of conductor (mm ²)	Nominal thickness (ti)
16	1.2
25	1.2
35	1.2
50	1.5
70	1.5
95	1.5

The smallest of measured value of thickness shall not fall below the nominal value(t_i) specified in above table by more than 0.1mm +0.1(t_i)

Upper surface of Cable insulation material shall be black and suitable to Ultra Violet Ray resistant, salt laden sprays, chemical pollution, ageing effects, abrasion and mechanical & electrical stress at temperature upto 90°C in normal operation and 250°C under short circuit condition. The Carbon black content in XLPE insulation shall be 2.5%±0.5%

8.1.2 DIMENSIONAL AND ELECTRICAL DATA

The standard sizes and technical characteristics for Single Core Power Cable is given below:

Nominal Cross sectional Area of Al. Conductor (sq. mm)	Maximum DC resistance at 20°C (ohm/km)	Minimum No. of Strand	Nominal Insulation Thickness (mm.)
16	1.91	6	1.2
25	1.2	6	1.2
35	0.868	6	1.2
50	0.641	6	1.5
70	0.443	12	1.5
95	0.32	15	1.5

Note : The resistance values given in the Table are the max. permissible one.

8.2 DETAILS OF MESSENGER-CUM-NEUTRAL WIRE :

The Messenger Conductor should be multi stranded Heat treated Aluminium Magnesium-Silicon Alloy Wires containing approximate 0.5% magnesium and 0.5% silicon conforming to IS-398 (Part-IV)/1984 **with latest amendment.**

The insulated messenger wire shall be of Stranded compacted Circular.

Messenger wire shall be suitably compacted to have smooth round surface to avoid damage to the other insulating sheath of phase conductor twisted around the messenger.

The technical characteristics of the messenger wire shall be as follows:

Nominal Cross sectional Area of Phase Conductor (sq. mm)	Nominal Area of Messenger Wire (sq. mm)	No. of Strand & Strand dia. (No./mm)	Dia. of Compacted Conductor (mm)	Maximum DC resistance at 20°C (ohm/km)
16	25	7/2 .13	5.64	1.38
25	25	7/2.13	5.64	1.38
35	25	7/2.13	5.64	1.38
50	35	7/2.52	6.68	0.986
70	50	7/3.02	7.98	0.689
95	70	7/3.57	9.44	0.492

8.3 SPLICING IN WIRES:

Splices in wires shall generally comply with the requirements of IEC 1089. The Aluminium Alloy rods may be spliced by cold pressure butt welding before drawing provided the manufacturer can guarantee that

the splice can develop 90% of the tensile strength of the spliced rod. Wires which breaks during stranding may be spliced by cold pressure butt welding provided that:
no two splices with completed conductor occur within 15 m of each other and no two splices in any individual wires are less than 150 m apart.

The finished splice shall be smooth and at no point shall the cross sectional area be less than that of un spliced wire. Splicing of the alloy wires on the stranding machine in order to utilise lengths of wires on reels shall not be permitted

8.4 DESIGNATION AND PARAMETERS OF FINISHED CABLES:

The designation and parameters of finished cables are given in the following table:

Sl. No.	Designation
1	LT AB Cable of size 3x16+1x25 sq. mm
2	LT AB Cable of size 3x25+1x25 sq. mm.
3	LT AB Cable of size 3x35+1x25 sq. mm
4	LT AB Cable of size 3X50+1X35 sq. mm.
5	LT AB Cable of size 3X50+1X16+1X35 sq. mm.
6	LT AB Cable of size 3X70+1X16+1X50 sq. mm
7	LT AB Cable of size 3x95+1x16 +1x70 sq. mm

9.0 LAYING UP :

Three Power Cores and one street lighting Core (if any) should be twisted over insulated messenger wire with right hand direction of lay. Lay ratio shall be as specified in IS:14255/1995 **with latest amendment**.

10.0 CORE IDENTIFICATION

Three Power Cores should be identified by Ridges one, two and three. **Insulated messenger** wire should be identified by four ridges. Street lighting Core (if any) should be without any ridges.

11.0 CABLE IDENTIFICATION

Cables with XLPE insulation shall be identified throughout the length of the Cable by the legend "XLPE 90" and year of manufacture by embossing.

12.0 TESTS:

All the cable sizes i.e. items offered should have been fully Type Tested as per the relevant standards at any NABL accredited Lab bearing an Accreditation body Logo/Central or State Govt. recognised test house or Laboratory.

12.1 TYPE TEST

The bidder shall furnish the type test reports conducted on similar type of cable as per IS & Tech –Spec along with the offer. These Type Tests must have been conducted within last five years prior to date of bid (Techno-Commercial) opening.

The following Type tests shall be carried out on the cables as per Relevant IS & Type tests certificates shall be furnished invariably with the offer.

1. Tests on Phase/Street Light Conductor :
 - i) Resistance Test.
2. Test on Messenger Conductor ;
 - i) Breaking Load.
 - ii) Elongation Test.
 - iii) Resistance Test.
3. Physical Test for XLPE insulation :
 - i) Tensile Strength and elongation at break.
 - ii) Ageing in air oven.
 - iii) Hot Set Test.
 - iv) Shrinkage Test.
 - v) Water Absorption (Gravimetric)
4. Test for thickness of insulation.
5. Insulation resistance (Volume resistivity) Test.
6. High Voltage Test.

Report of above and any other type test provided in the relevant Indian Standards shall also be supplied.

12.2 ROUTINE TEST :

These shall be carried out on all sizes of all drums of un armoured Cables by the supplier in presence of the purchaser's representative if so desired by the purchaser and shall be got approved before despatch.

The Cable shall be inspected and tested at the manufacturer's works before despatch as per ISS mentioned in schedule of requirement. The manufacturer shall arrange all the necessary machinery, apparatus and labour required for the testing purpose.

The routine tests embodied in the respective ISS are as per list given below :

1. Conductor Resistance Test.
2. High Voltage Test.

Any other routine test provided in the relevant Indian Standards shall also got conducted by the supplier.

12.3 ACCEPTANCE TEST :

These tests as per relevant IS & IEC shall be carried out in the presence of the purchaser's representative if so desired by the purchaser. The following shall constitute the acceptance tests.

- i) Breaking Load Test for messenger Conductor.
- ii) Elongation test for messenger conductor.
- iii) Conductor resistance test for both messenger and XLPE conductor.
- iv) Test for thickness of insulation of XLPE conductor.
- v) Tensile Strength and Elongation at Break Test for both messenger and XLPE conductor.
- vi) Hot Set Test for XLPE insulation.
- vii) Insulation resistance test for XLPE conductor
- viii) High Voltage Test for XLPE conductor

13.0 SEQUENTIAL MARKING :

Marking of the sequential length shall be provided on any one of phase cores by printing/ engraving. Weight of cable need not be provided on the core.

Each meter length shall be embossed with the trade name of manufacturer and the word 'ELECTRIC-WBSEDCL', ISI mark.

14.0 PACKING & MARKING :

The Cable shall be supplied on suitable sized wooden non-returnable drum of robust construction for each length of Cables as per IS: 10418. Manufacture will provide the Multiplying Factor considering the Lay Ratio of the cable to measure the length of cable in a drum. **Multiplying factor would be provided in the GTP as well as in the drum.** Sequential length of the core & the declared cable length shall be marked on the cable drum. Net weight & Gross weight shall be provided on the drum.

A layer of water proof paper shall be applied to the surface of the drums and over the outer cable layer. A clear space at least 40 mm. shall be left between the Cable and Logging. Wood preservative shall be applied to the entire drum. Packing shall be sturdy to protect the cable from any injury during transportation handling and storage. Both Cable ends shall be sealed with PVC/Rubber Caps to eliminate ingress of moisture. Each Cable Drum shall have following information stencilled on it.

- a) Manufacturer's Name or Trade Mark.
- b) Year of Manufacture.
- c) Nominal Cross-Sectional area of the Cable Conductor.
- d) Type of Cable and Voltage Grade.
- e) Length of the Cable on the Drum.
- f) Number of the Cores.
- g) ISI Certification Mark, IS reference.
- h) No. of lengths on the Drum (if more than one)
- i) Direction of rotation of drum (by means of an arrow)
- j) Approximate gross weight in Kg./Km.

k) Multiplying Factor

The standard Drum Length will be 500 meters in each Drum for all sizes subject to a tolerance of $\pm 5\%$ **and overall quantity tolerance should be -1%.**

15. INSPECTION /WITNESSING OF TESTS:

The Cable shall be inspected at manufacturer's works before despatch as per IS-7098 (Part-I)/2011 (**with latest amendments**) & IS 14255/1995 **with latest amendments**. All the acceptance tests embodied in the above shall be performed by the Inspecting Officer. The Manufacturer shall arrange without making any extra charges with all the necessary machinery, apparatus and labour requirement for the testing purpose. The Cable requirement for testing shall be to Firm's Account.

6 copies of type test certificate lot wise for each type of cable shall be sent to the purchaser for acceptance. Type test certificate for each lot and routine test certificate for each drum of cables shall be submitted to purchaser for approval before despatch of cables from the works. The test certificate shall be completed with all results.

16.0 LITERATURE AND MANUAL:

To be submitted as per General Terms & Conditions of Contract.

17.0 GUARANTEED TECHNICAL PARTICULARS:

To be submitted along with Tender documents in triplicate as per format annexed herewith. Technical characteristic shall be guaranteed by the bidder. In case of failure of materials to meet the guarantee, WBSEDCL shall have right to reject the material.

GTP for LT AB CABLE

(To be filled and signed up by each bidder or manufacturer)

1.0	Manufacturer's Name & Address		
2.0	AB Cable Sizes		
3.0	Lists of Standard applicable.		IS: 14255-1995, IS: 8130-1984, IS 398 (Part-4)-1994, IS-10418-1982 with latest Amendment
4.A	Power Core		
	a)	No of Core	
	b)	Nominal cross sectional area of Power Core	
	c)	No. of strands of Power Core	
	d)	Strand dia of Power core	
	e)	Dia. of Compacted Conductor (MM) of Power Core	
	f)	Insulation Nominal Thickness	
	g)	Insulation Minimum Thickness	
	h)	Approximate dia over insulation	
	i)	Continuous current carrying capacity of Cable in Air at Ambient Temp. 45 °C (Amp.)	
	j)	Identification of Power Cores	The phase conductors shall be provided with one, two or three ridges for quick identification
	k)	Max. DC resistance of conductor at 200C(ohm/km)	
	l)	Approx. Mass (Kg/KM)	
4.B	Neutral-Cum-Messenger Core		
	a)	No of Core	
	b)	Nominal cross sectional area of Neutral-Cum-Messenger Core	
	c)	No. of strands of Neutral-Cum-Messenger Core	
	d)	Strand dia. Neutral-Cum-Messenger Core	
	e)	Dia. of Compacted Conductor (mm) of Neutral-Cum-Messenger Core	
	f)	Insulation Nominal Thickness	
	g)	Insulation Minimum Thickness	
	h)	Approximate dia. over insulation	
	i)	Continuous current carrying capacity of Messenger Core in Air at Ambient Temp. 45°C (Amp.)	
	j)	Identification of Neutral-Cum- Messenger Core	Insulated Messenger wire shall have four ridges as per IS 14255:1995.
	k)	Approx. breaking load (KN)	
	l)	Max. DC resistance of conductor at 200C	
	m)	Approx. Mass (Kg/KM)	

4.C	Street Lighting Core				
	a)	No of Core			
	b)	Nominal cross sectional area of Street Lighting Core			
	c)	No. of strands			
	d)	Strand dia			
	e)	Dia. of Compacted Conductor (MM)			
	f)	Insulation Nominal Thickness			
	g)	Insulation Minimum Thickness			
	h)	Approximate dia over insulation			
	i)	Continuous current carrying capacity of Street Lighting Core in Air at Ambient Temp. 450C (Amp.)			
	j)	Identification of Street Lighting Core.	The insulated street light conductor shall not have any identification mark		
	k)	Max. DC resistance of conductor at 200C			
	l)	Approx. Mass (Kg/KM)			
5.0	Details Of Power Core Conductor & Street Light Core Conductor				
	a)	Material	Aluminium conductor shall be H4 grade complying with the requirements of IS-8130-1984 with up to date amendments.		
	b)	Flexibility Class as per IS : 8130-1984	Class-2		
	c)	Form of Conductor.	Stranded compacted circular		
	d)	Maximum Continuous Conductor Temperature (⁰ C)	90 ⁰ C		
	e)	Maximum Short time Conductor Temperature (⁰ C)	250 ⁰ C		
6.0	Details Of Neutral-Cum-Messenger Core Conductor				
	a)	Material	Heat treated Aluminium Magnesium-Silicon Alloy Wires containing approximate 0.5% magnesium and 0.5% silicon conforming to IS-398 (Part-IV)/1984 with up to date amendments.		
	b)	Form of Conductor.	Stranded Compacted Circular		
7.0	Insulation				
	a)	Material.	Cross linked polyethylene		
	b)	Method of application of Insulation	Extrusion		
	c)	Type of curing of XLPE Insulation Completed Cable.	Curing with Sioplus Technology		
	d)	Colour of Insulation	Black		
8.0	Voltage Grade of Cable.		1100V	1100V	1100V
	a)	Service Voltage.	1100 V	1100 V	1100V
	b)	Neutral to Phase voltage	635 V	635 V	635 V

9.0	De-rating factor: De-rating factors for variation in Air Temp.			
	Air Temp. 0C			
10.	Direction of Laying	Right Hand Direction		
11.	Details about Completed Cable :			
a)	Approx. overall diameter(mm).			
b)	Approx. Weight Kg./KM			
c)	Allowable sag as a percentage of span length at 40C ambient temperature			
12.	Details about Cable Drums :			
a)	Standard drum length & tolerance of each drum	500 mtrs. +5%		
b)	Dimension of Drum	As per IS:10418-1982		
c)	Shipping Weight.			
13.	Overall quantity tolerance	-1%		
14.	Sequential Marking	Marking of the sequential length shall be provided on any one of phase cores by printing/ engraving.		
15.	Embossing	Each metre length shall be embossed with the Trade Name of Manufacturer and the word 'ELECTRIC WBSEDCL'.		
16.	Cable identification	Cables with XLPE insulation shall be identified throughout the length of the Cable by the legend "XLPE 90" and year of manufacture by embossing.		
17.	Multiplying Factor considering Lay Ratio of cable to Measure the Length of cable in Drum			

Signature with designation & seal
with Name of the Firm